

P8131 Homework 1

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Problem 1

Show that the following distributions belong to the exponential family. Find the natural parameter θ , scale parameter ϕ and convex function $b(\theta)$. Also find the $E(Y)$ and $\text{Var}(Y)$ as functions of the natural parameter. Specify the canonical link functions.

The exponential family (with dispersion) has the form:

$$f(y|\theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

Properties: $E(Y) = b'(\theta)$, $\text{Var}(Y) = \phi b''(\theta)$.

The canonical link function is $g = (b')^{-1}$, which satisfies $g(\mu) = \theta$.

(a) Exponential Distribution $\text{Exp}(\lambda)$

$$f(y; \lambda) = \lambda e^{-\lambda y}, \quad y > 0$$

Rewrite:

$$f(y; \lambda) = \exp\{\log \lambda - \lambda y\} = \exp\{-\lambda y - (-\log \lambda)\}$$

Comparing with the exponential family form:

- **Natural parameter:** $\theta = -\lambda$ (so $\lambda = -\theta$)
- **Dispersion parameter:** $\phi = 1$
- **Convex function:** $b(\theta) = -\log(-\theta)$

Verification:

- $b'(\theta) = -\frac{1}{\theta} = \frac{1}{\lambda}$
- $b''(\theta) = \frac{1}{\theta^2} = \frac{1}{\lambda^2}$

Mean and Variance:

- $E(Y) = b'(\theta) = -\frac{1}{\theta} = \frac{1}{\lambda}$
- $\text{Var}(Y) = \phi b''(\theta) = \frac{1}{\theta^2} = \frac{1}{\lambda^2}$

Canonical link function: $g(\mu) = \theta = -\lambda = -\frac{1}{\mu}$ (negative reciprocal link)

(b) Binomial Distribution $\text{Bin}(n, \pi)$, n known

$$f(y; \pi) = \binom{n}{y} \pi^y (1 - \pi)^{n-y}, \quad y = 0, 1, \dots, n$$

Rewrite:

$$\begin{aligned} f(y; \pi) &= \binom{n}{y} \exp \{y \log \pi + (n - y) \log(1 - \pi)\} \\ &= \binom{n}{y} \exp \left\{ y \log \frac{\pi}{1 - \pi} + n \log(1 - \pi) \right\} \end{aligned}$$

Comparing with the exponential family form:

- **Natural parameter:** $\theta = \log \frac{\pi}{1 - \pi}$ (logit of π), so $\pi = \frac{e^\theta}{1 + e^\theta}$
- **Dispersion parameter:** $\phi = 1$
- **Convex function:** $b(\theta) = n \log(1 + e^\theta)$

Verification:

- $b'(\theta) = \frac{ne^\theta}{1 + e^\theta} = n\pi$
- $b''(\theta) = \frac{ne^\theta}{(1 + e^\theta)^2} = n\pi(1 - \pi)$

Mean and Variance:

- $E(Y) = b'(\theta) = n\pi = \frac{ne^\theta}{1 + e^\theta}$
- $\text{Var}(Y) = \phi b''(\theta) = n\pi(1 - \pi) = \frac{ne^\theta}{(1 + e^\theta)^2}$

Canonical link function: $g(\mu) = \log \frac{\mu/n}{1 - \mu/n} = \log \frac{\mu}{n - \mu}$ (logit link)

Note: If $Y^ = Y/n$ represents the proportion, then $g(\mu^*) = \log \frac{\mu^*}{1 - \mu^*}$.*

(c) Poisson Distribution $\text{Pois}(\lambda)$

$$f(y; \lambda) = \frac{1}{y!} \lambda^y e^{-\lambda}, \quad y = 0, 1, 2, \dots$$

Rewrite:

$$f(y; \lambda) = \frac{1}{y!} \exp\{y \log \lambda - \lambda\}$$

Comparing with the exponential family form:

- **Natural parameter:** $\theta = \log \lambda$, so $\lambda = e^\theta$
- **Dispersion parameter:** $\phi = 1$
- **Convex function:** $b(\theta) = e^\theta$

Verification:

- $b'(\theta) = e^\theta = \lambda$
- $b''(\theta) = e^\theta = \lambda$

Mean and Variance:

- $E(Y) = b'(\theta) = e^\theta = \lambda$
- $\text{Var}(Y) = \phi b''(\theta) = e^\theta = \lambda$

Canonical link function: $g(\mu) = \log \mu$ (log link)

(d) Negative Binomial Distribution $\text{NB}(m, \beta)$, m known

$$f(y; \beta) = \binom{y+m-1}{m-1} \beta^m (1-\beta)^y, \quad y = 0, 1, 2, \dots$$

Rewrite:

$$f(y; \beta) = \binom{y+m-1}{m-1} \exp\{m \log \beta + y \log(1-\beta)\}$$

Comparing with the exponential family form:

- **Natural parameter:** $\theta = \log(1-\beta)$, so $1-\beta = e^\theta$ and $\beta = 1 - e^\theta$
- **Dispersion parameter:** $\phi = 1$
- **Convex function:** $b(\theta) = -m \log(1 - e^\theta) = -m \log \beta$

Verification:

- $b'(\theta) = \frac{me^\theta}{1-e^\theta} = \frac{m(1-\beta)}{\beta}$
- $b''(\theta) = \frac{me^\theta}{(1-e^\theta)^2} = \frac{m(1-\beta)}{\beta^2}$

Mean and Variance:

- $E(Y) = b'(\theta) = \frac{m(1-\beta)}{\beta} = \frac{me^\theta}{1-e^\theta}$
- $\text{Var}(Y) = \phi b''(\theta) = \frac{m(1-\beta)}{\beta^2} = \frac{me^\theta}{(1-e^\theta)^2}$

Canonical link function: $g(\mu) = \log\left(\frac{\mu}{\mu+m}\right)$ (derived from $\theta = \log(1-\beta) = \log \frac{\mu}{\mu+m}$)

(e) Gamma Distribution $\text{Gamma}(\alpha, \beta)$, α known

$$f(y; \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}, \quad y > 0$$

Rewrite:

$$f(y; \beta) = \exp\{-\beta y + \alpha \log \beta + (\alpha - 1) \log y - \log \Gamma(\alpha)\}$$

Comparing with the exponential family form:

- **Natural parameter:** $\theta = -\beta$ (so $\beta = -\theta$)
- **Dispersion parameter:** $\phi = 1/\alpha$
- **Convex function:** $b(\theta) = -\log(-\theta)$

To match the standard form with dispersion:

$$f(y|\theta, \phi) = \exp\left\{\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)\right\}$$

We have $\phi = 1/\alpha$, $b(\theta) = -\log(-\theta)$, where $c(y, \phi) = (\frac{1}{\phi} - 1) \log y - \log \Gamma(1/\phi) - \frac{1}{\phi} \log \phi$.

Verification:

- $b'(\theta) = -\frac{1}{\theta} = \frac{1}{\beta}$
- $b''(\theta) = \frac{1}{\theta^2} = \frac{1}{\beta^2}$

Mean and Variance:

- $E(Y) = b'(\theta) = \frac{1}{\beta} = -\frac{1}{\theta} = \frac{\alpha}{\beta}$ (when $\phi = 1/\alpha$)
- $\text{Var}(Y) = \phi b''(\theta) = \frac{1}{\alpha} \cdot \frac{1}{\beta^2} = \frac{\alpha}{\beta^2}$

Canonical link function: $g(\mu) = -\frac{1}{\mu}$ (negative reciprocal link)

Problem 2

Assume $Y_1, Y_2, \dots, Y_n$ are independent and follow a binomial distribution where $Y_i \sim \text{Bin}(m, \pi_i)$ and m is known. Furthermore, assume $\log \frac{\pi_i}{1-\pi_i} = X_i \beta$. What are the expressions of deviance residuals and Pearson residuals respectively (use $\hat{\beta}$ to represent the MLE)? What are the expressions of the deviance and Pearson's χ^2 statistic?

Let $\hat{\pi}_i = \frac{e^{X_i \hat{\beta}}}{1+e^{X_i \hat{\beta}}}$ be the fitted probability, where $\hat{\beta}$ is the MLE.

Pearson Residuals

The Pearson residual is defined as:

$$r_{P_i} = \frac{Y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$$

For binomial data with $\hat{\mu}_i = m\hat{\pi}_i$ and $V(\hat{\mu}_i) = m\hat{\pi}_i(1 - \hat{\pi}_i)$:

$$r_{P_i} = \frac{Y_i - m\hat{\pi}_i}{\sqrt{m\hat{\pi}_i(1 - \hat{\pi}_i)}}$$

Pearson's χ^2 Statistic

$$G = \sum_{i=1}^n r_{P_i}^2 = \sum_{i=1}^n \frac{(Y_i - m\hat{\pi}_i)^2}{m\hat{\pi}_i(1 - \hat{\pi}_i)}$$

Under the assumed model, $G \stackrel{\text{approx}}{\sim} \chi^2(n-p)$ for large m .

Deviance Residuals

The deviance residual is defined as:

$$r_{D_i} = \text{sign}(Y_i - \hat{\mu}_i) \sqrt{d_i}$$

where $d_i = D(Y_i, \hat{\mu}_i)$ is the individual contribution to the deviance.

For binomial data, the deviance contribution is:

$$d_i = 2 \left[Y_i \log \frac{Y_i}{m\hat{\pi}_i} + (m - Y_i) \log \frac{m - Y_i}{m(1 - \hat{\pi}_i)} \right]$$

Therefore:

$$r_{D_i} = \text{sign}(Y_i - m\hat{\pi}_i) \sqrt{2 \left[Y_i \log \frac{Y_i}{m\hat{\pi}_i} + (m - Y_i) \log \frac{m - Y_i}{m(1 - \hat{\pi}_i)} \right]}$$

Note: The convention $0 \cdot \log 0 = 0$ is used when $Y_i = 0$ or $Y_i = m$.

Deviance

$$D = \sum_{i=1}^n d_i = 2 \sum_{i=1}^n \left[Y_i \log \frac{Y_i}{m\hat{\pi}_i} + (m - Y_i) \log \frac{m - Y_i}{m(1 - \hat{\pi}_i)} \right]$$

Under the assumed model, $D \stackrel{\text{approx}}{\sim} \chi^2(n-p)$ for large m .

Problem 3

Consider the binary response variable $Y \sim \text{Bernoulli}$ with $P(Y = 1) = \pi$ and $P(Y = 0) = 1 - \pi$. Observations Y_i , $i = 1, \dots, n$, are independent and identically distributed as Y .

(a) Test Statistics for $H_0 : \pi = \pi_0$

The log-likelihood function is:

$$\ell(\pi) = \sum_{i=1}^n \{Y_i \log \pi + (1 - Y_i) \log(1 - \pi)\} = n\bar{Y} \log \pi + n(1 - \bar{Y}) \log(1 - \pi)$$

where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$.

Score function:

$$s(\pi) = \frac{\partial \ell}{\partial \pi} = \frac{n\bar{Y}}{\pi} - \frac{n(1 - \bar{Y})}{1 - \pi} = \frac{n(\bar{Y} - \pi)}{\pi(1 - \pi)}$$

Fisher information:

$$\iota(\pi) = E \left[-\frac{\partial^2 \ell}{\partial \pi^2} \right] = \frac{n}{\pi(1 - \pi)}$$

MLE: $\hat{\pi} = \bar{Y}$

Wald Test Statistic

$$TS_W = (\hat{\pi} - \pi_0)^T \iota(\hat{\pi}) (\hat{\pi} - \pi_0) = \frac{n(\bar{Y} - \pi_0)^2}{\bar{Y}(1 - \bar{Y})}$$

Score Test Statistic

$$TS_S = s(\pi_0)^T \iota(\pi_0)^{-1} s(\pi_0) = \frac{n(\bar{Y} - \pi_0)^2}{\pi_0(1 - \pi_0)}$$

Likelihood Ratio Test Statistic

$$TS_{LR} = 2\{\ell(\hat{\pi}) - \ell(\pi_0)\} = 2n \left[\bar{Y} \log \frac{\bar{Y}}{\pi_0} + (1 - \bar{Y}) \log \frac{1 - \bar{Y}}{1 - \pi_0} \right]$$

Under H_0 , all three statistics asymptotically follow $\chi^2(1)$.

(b) Numerical Calculations

Data: $(0, 1, 0, 0, 1, 0, 0, 0, 1, 0)$, so $n = 10$, $\sum Y_i = 3$, $\bar{Y} = 0.3$.

```

# Data
n <- 10
y_bar <- 0.3

# Function to calculate test statistics
calc_tests <- function(pi0, y_bar, n) {
  # Wald test
  if (y_bar == 0 | y_bar == 1) {
    wald <- NA
  } else {
    wald <- n * (y_bar - pi0)^2 / (y_bar * (1 - y_bar))
  }

  # Score test
  score <- n * (y_bar - pi0)^2 / (pi0 * (1 - pi0))

  # LRT
  if (y_bar == 0) {
    lrt <- 2 * n * log((1 - y_bar) / (1 - pi0))
  } else if (y_bar == 1) {
    lrt <- 2 * n * log(y_bar / pi0)
  } else {
    lrt <- 2 * n * (y_bar * log(y_bar / pi0) + (1 - y_bar) * log((1 - y_bar) / (1 - pi0)))
  }

  # p-values
  p_wald <- 1 - pchisq(wald, 1)
  p_score <- 1 - pchisq(score, 1)
  p_lrt <- 1 - pchisq(lrt, 1)

  return(c(Wald = wald, Score = score, LRT = lrt,
           p_Wald = p_wald, p_Score = p_score, p_LRT = p_lrt))
}

# Critical value
alpha <- 0.05
crit_val <- qchisq(1 - alpha, 1)
cat("Chi-squared critical value (alpha = 0.05):", round(crit_val, 4), "\n\n")

```

```
## Chi-squared critical value (alpha = 0.05): 3.8415
```

```

# (i) pi0 = 0.1
cat("(i) H0: pi = 0.1\n")

```

```
## (i) H0: pi = 0.1
```

```

res1 <- calc_tests(0.1, y_bar, n)
print(round(res1, 4))

```

```

##      Wald   Score    LRT  p_Wald p_Score  p_LRT
##  1.9048  4.4444  3.0733  0.1675  0.0350  0.0796

```

```
# (ii) pi0 = 0.3
cat("\n(ii) H0: pi = 0.3\n")
```

```
##
## (ii) H0: pi = 0.3
```

```
res2 <- calc_tests(0.3, y_bar, n)
print(round(res2, 4))
```

```
##      Wald      Score      LRT  p_Wald p_Score  p_LRT
##      0         0         0         1       1       1
```

```
# (iii) pi0 = 0.5
cat("\n(iii) H0: pi = 0.5\n")
```

```
##
## (iii) H0: pi = 0.5
```

```
res3 <- calc_tests(0.5, y_bar, n)
print(round(res3, 4))
```

```
##      Wald      Score      LRT  p_Wald p_Score  p_LRT
##  1.9048  1.6000  1.6457  0.1675  0.2059  0.1996
```

(c) Comparison of Tests

```
results <- data.frame(
  pi0 = c(0.1, 0.3, 0.5),
  Wald = c(res1[1], res2[1], res3[1]),
  Score = c(res1[2], res2[2], res3[2]),
  LRT = c(res1[3], res2[3], res3[3]),
  Wald_decision = ifelse(c(res1[1], res2[1], res3[1]) > crit_val, "Reject", "FTR"),
  Score_decision = ifelse(c(res1[2], res2[2], res3[2]) > crit_val, "Reject", "FTR"),
  LRT_decision = ifelse(c(res1[3], res2[3], res3[3]) > crit_val, "Reject", "FTR")
)
knitr::kable(results, digits = 4, caption = "Test Statistics and Decisions")
```

Table 1: Test Statistics and Decisions

pi0	Wald	Score	LRT	Wald_decision	Score_decision	LRT_decision
0.1	1.9048	4.4444	3.0733	FTR	Reject	FTR
0.3	0.0000	0.0000	0.0000	FTR	FTR	FTR
0.5	1.9048	1.6000	1.6457	FTR	FTR	FTR

Conclusions:

- For $\pi_0 = 0.1$: Tests disagree—Score rejects while Wald and LRT fail to reject.
- For $\pi_0 = 0.3$: All tests fail to reject (expected since $\hat{\pi} = \pi_0$).
- For $\pi_0 = 0.5$: All tests fail to reject.

With only $n = 10$, statistical power is low—even though $\hat{\pi} = 0.3$, we lack sufficient evidence to reject $\pi_0 = 0.1$ (for Wald and LRT) or $\pi_0 = 0.5$ (for all tests). Asymptotically (large n), all three tests are equivalent and would have greater power to detect such differences.

Problem 4

Assume $Y_i \sim \text{Pois}(\lambda)$, $i = 1, \dots, n$. We are interested in testing $H_0 : \log \lambda = \log \lambda_0$.

Since $\log$ is a monotonic function, $H_0 : \log \lambda = \log \lambda_0 \Leftrightarrow H_0 : \lambda = \lambda_0$. Thus the test statistics are identical regardless of parameterization. These results are the same as Lecture 4, slide 10-11.

Log-likelihood:

$$\ell(\lambda) = \sum_{i=1}^n \{Y_i \log \lambda - \lambda - \log Y_i!\} = n\bar{Y} \log \lambda - n\lambda - \sum_{i=1}^n \log Y_i!$$

Score function:

$$s(\lambda) = \frac{\partial \ell}{\partial \lambda} = \frac{n\bar{Y}}{\lambda} - n = \frac{n(\bar{Y} - \lambda)}{\lambda}$$

Fisher information:

$$\iota(\lambda) = E \left[-\frac{\partial^2 \ell}{\partial \lambda^2} \right] = E \left[\frac{n\bar{Y}}{\lambda^2} \right] = \frac{n}{\lambda}$$

MLE: $\hat{\lambda} = \bar{Y}$

Wald Test Statistic

$$TS_W = (\hat{\lambda} - \lambda_0)^T \iota(\hat{\lambda}) (\hat{\lambda} - \lambda_0) = \frac{n(\bar{Y} - \lambda_0)^2}{\bar{Y}}$$

Score Test Statistic

$$TS_S = s(\lambda_0)^T \iota(\lambda_0)^{-1} s(\lambda_0) = \frac{[n(\bar{Y} - \lambda_0)/\lambda_0]^2}{n/\lambda_0} = \frac{n(\bar{Y} - \lambda_0)^2}{\lambda_0}$$

$$TS_S = \frac{n(\bar{Y} - \lambda_0)^2}{\lambda_0}$$

Likelihood Ratio Test Statistic

$$TS_{LR} = 2\{\ell(\hat{\lambda}) - \ell(\lambda_0)\} = 2\{n\bar{Y} \log \bar{Y} - n\bar{Y} - n\bar{Y} \log \lambda_0 + n\lambda_0\}$$

$$TS_{LR} = 2n \left\{ \bar{Y} \log \frac{\bar{Y}}{\lambda_0} - (\bar{Y} - \lambda_0) \right\}$$

Under $H_0 : \lambda = \lambda_0$, all three test statistics asymptotically follow $\chi^2(1)$.